Higher National Diploma in Engineering (Electrical)
Fourth Year, First Semester Examination – 2018
EE 3206 – Robotics & Automation

Instructions for Candidates:
Answer **any Five (05)** questions.
All question carry equal marks

No. of questions : 06
No. of pages : 02
Time : **03 hours**

Q1.

1. Briefly explain the manufacturing of silicon wafer? (6 marks)

2. Write the short notes on (6 marks)

   (i) Doping

   (ii) Oxidation

3. Explain the Czochralski (CZ) Silicon crystal growing method? (8 marks)

Q2.

1. Briefly explain the mean time to failure (MTTF)? (4 mark)

2. What are the advantages of a using printed circuit board (PCB)? (8 marks)

3. What are the Advantages of Surface Mount Technology? (8 marks)

Q3.

1. Define the term of Degree of freedom? (4 marks)
2. Write the short notes on Accuracy and Repeatability? (8 marks)
3. Write five reasons to use of robotics in manufacturing? (8 marks)

Q4.

1. Briefly explain the Gear and belts in robot mechanical transmission system? (6 marks)
2. Briefly explain the Proximity Sensor? (6 marks)
3. List out three advantages of absolute encoder and three advantages of incremental encoders? (8 marks)

Q5.

1. Explain the tech methods of programming in industrial robots? (6 marks)
2. Write the short notes on (6 marks)
   (i) Revolute Joint
   (ii)Prismatic Joint
3. Briefly explain the fixed automation and Programmable automation? (8 marks)

Q6.

1. Define the term of Robot cell? (4 marks)
2. Briefly explain In- line robot work cell and Robot-centered work cell? (8 marks)
3. Briefly Explain Cartesian coordinate robot and cylindrical robot? (8 marks)